

Classified as hazardous in accordance with the criteria of EPA New Zealand

## Section 1 - Identification

### Product Identifier

|                      |                                  |
|----------------------|----------------------------------|
| Product Name         | <u>Tetrahydro-4H-pyran-4-one</u> |
| CAS No               | 29943-42-8                       |
| Synonyms             | Tetrahydro-2H-pyran-4-one.       |
| Molecular Formula    | C5 H8 O2                         |
| Molecular Weight     | 100.12                           |
| Recommended Use      | Laboratory chemicals.            |
| Uses advised against | No Information available         |

|                         |                                                                                             |
|-------------------------|---------------------------------------------------------------------------------------------|
| Product Code            | <b>L06913</b>                                                                               |
| Address                 | Thermo Fisher Scientific New Zealand Ltd<br>244 Bush Road, Albany,<br>Auckland, New Zealand |
| Emergency Tel.          | <b>CHEMTREC®</b><br><b>09 980 6780 or +64 9 980 6780</b>                                    |
| Telephone / Fax Numbers | Tel: 09 980 6700<br>Fax: 09 980 6788                                                        |
| E-mail address          | <u>ANZinfo@thermofisher.com</u>                                                             |

## Section 2 - Hazard(s) Identification

### Classification under Work Safe New Zealand

Classified as hazardous in accordance with the criteria of EPA New Zealand

### GHS Classification

#### Physical hazards

Flammable liquids

Category 3

#### Health hazards

Based on available data, the classification criteria are not met

#### Environmental hazards

Based on available data, the classification criteria are not met

### Label Elements

**Signal Word****Warning****Hazard Statements**

H226 - Flammable liquid and vapor

**Precautionary Statements****Prevention**

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking

P233 - Keep container tightly closed

P240 - Ground and bond container and receiving equipment

P242 - Use non-sparking tools

P243 - Take action to prevent static discharges

P280 - Wear protective gloves/protective clothing/eye protection/face protection

**Response**

P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower

P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish

**Storage**

P403 + P235 - Store in a well-ventilated place. Keep cool

**Disposal**

P501 - Dispose of contents/ container to an approved waste disposal plant

**Other hazards which do not result in classification**

## Section 3 - Composition and Information on Ingredients

| Component                 | CAS No     | Weight % |
|---------------------------|------------|----------|
| Tetrahydro-4H-pyran-4-one | 29943-42-8 | > 95     |

## Section 4 - First Aid Measures

**Description of first aid measures**

|                                           |                                                                                                                   |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                     | If symptoms persist, call a physician.                                                                            |
| <b>New Zealand Emergency Tel.</b>         | CHEMTREC®<br>09 980 6780 or +64 9 980 6780                                                                        |
| <b>Inhalation</b>                         | Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.      |
| <b>Eye Contact</b>                        | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.   |
| <b>Skin Contact</b>                       | Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician. |
| <b>Ingestion</b>                          | Clean mouth with water and drink afterwards plenty of water.                                                      |
| <b>Self-Protection of the First Aider</b> | No special precautions required.                                                                                  |

**First Aid Facilities**

Eyewash, safety shower and washroom.

**Most important symptoms and effects**

. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting

**Notes to Physician**

Treat symptomatically.

## **Section 5 - Fire Fighting Measures**

**Suitable Extinguishing Media**Water spray. Carbon dioxide (CO<sub>2</sub>). Dry chemical. Chemical foam. Water mist may be used to cool closed containers.**Extinguishing media which must not be used for safety reasons**

No information available.

**Specific Hazards Arising from the Chemical**

Combustible material. Vapors may travel to source of ignition and flash back. Flammable. Containers may explode when heated. Vapors may form explosive mixtures with air.

**Hazardous Combustion Products**Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>).**Special protective equipment and precautions for fire fighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## **Section 6 - Accidental Release Measures**

**Personal Precautions, Protective Equipment and Emergency Procedures****Emergency procedures**

Ensure adequate ventilation. Use personal protective equipment as required. Remove all sources of ignition. Take precautionary measures against static discharges.

**Environmental Precautions**

Should not be released into the environment.

**Methods for Containment and Clean Up**

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.

**Precautions to prevent secondary hazards**

Clean contaminated objects and areas thoroughly observing environmental regulations

**Reference to Other Sections**

Refer to protective measures listed in Sections 8 and 13.

## **Section 7 - Handling and Storage**

**Precautions for Safe Handling****Advice on safe handling**

Wear personal protective equipment/face protection. Ensure adequate ventilation. Avoid ingestion and inhalation. Do not get in eyes, on skin, or on clothing. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. Take precautionary measures against static discharges.

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

**Conditions for Safe Storage, Including any Incompatibilities****Storage Conditions**

Keep container tightly closed in a dry and well-ventilated place. Keep away from heat, sparks and flame.

**Incompatible Materials**

Strong oxidizing agents. Strong bases.

AS/NZS 2243.10:2004, Safety in laboratories - Storage of chemicals

AS 1940-2004 - The storage and handling of flammable and combustible liquids

## **Section 8 - Exposure Controls and Personal Protection**

**Control parameters****Exposure limits**

The product does not contain any hazardous materials with occupational exposure limits established.

**Biological limit values**

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

**Appropriate engineering controls****Engineering Measures**

Ensure adequate ventilation, especially in confined areas. Use explosion-proof electrical/ventilating/lighting equipment. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

**Individual protection measures, such as personal protective equipment****Eye Protection**

Wear safety glasses with side shields (or goggles) (Australian/New Zealand Standard AS/NZS 1337 - Eye protectors for Industrial applications)

**Hand Protection**

Protective gloves

| Glove material                                    | Breakthrough time                    | Glove thickness | AUS/NZ Standard | Glove comments        |
|---------------------------------------------------|--------------------------------------|-----------------|-----------------|-----------------------|
| Nitrile rubber, Neoprene,<br>Natural rubber, PVC. | See manufacturers<br>recommendations | -               | AS/NZS 2161     | (minimum requirement) |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

**Skin and body protection**

Long sleeved clothing

**Respiratory Protection**

Use an AS/NZS 1716 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained in line with AS/NZS 1715 on the use and maintenance of respiratory protective devices (or AUS/NZ equivalent)

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice.

**Environmental exposure controls**

No information available.

## Section 9 - Physical and Chemical Properties

### Information on basic physical and chemical properties

|                                         |                                 |                                   |
|-----------------------------------------|---------------------------------|-----------------------------------|
| Physical State                          | Liquid                          |                                   |
| Appearance                              | Light yellow                    |                                   |
| Odor                                    | No information available        |                                   |
| Odor Threshold                          | No data available               |                                   |
| pH                                      | No information available        |                                   |
| Melting Point/Range                     | No data available               |                                   |
| Softening Point                         | No data available               |                                   |
| Boiling Point/Range                     | 166 - 166 °C / 330.8 - 330.8 °F | @ 760 mmHg                        |
| Flammability (liquid)                   | Flammable                       | On basis of test data             |
| Flammability (solid,gas)                | Not applicable                  | Liquid                            |
| Explosion Limits                        | No data available               |                                   |
| Flash Point                             | 56 °C / 132.8 °F                | Method - No information available |
| Autoignition Temperature                | No data available               |                                   |
| Decomposition Temperature               | No data available               |                                   |
| Viscosity                               | No data available               |                                   |
| Water Solubility                        | Miscible                        |                                   |
| Solubility in other solvents            | No information available        |                                   |
| Partition Coefficient (n-octanol/water) |                                 |                                   |
| Vapor Pressure                          | No data available               |                                   |
| Density / Specific Gravity              | 1.080                           |                                   |
| Bulk Density                            | Not applicable                  | Liquid                            |
| Vapor Density                           | 3.45                            | (Air = 1.0)                       |
| Particle characteristics                | Not applicable (liquid)         |                                   |

### Other information

|                      |                                        |
|----------------------|----------------------------------------|
| Molecular Formula    | C5 H8 O2                               |
| Molecular Weight     | 100.12                                 |
| Explosive Properties | explosive air/vapour mixtures possible |

## Section 10 - Stability and Reactivity

|                                  |                                                                                          |
|----------------------------------|------------------------------------------------------------------------------------------|
| Reactivity                       | None known, based on information available                                               |
| Stability                        | No information available.                                                                |
| Sensitivity to Mechanical Impact | No information available                                                                 |
| Sensitivity to Static Discharge  | No information available                                                                 |
| Hazardous Polymerization         | No information available.                                                                |
| Hazardous Reactions              | No information available.                                                                |
| Conditions to Avoid              | Incompatible products, Keep away from open flames, hot surfaces and sources of ignition. |
| Incompatible Materials           | Strong oxidizing agents, Strong bases.                                                   |
| Hazardous Decomposition Products | Carbon monoxide (CO). Carbon dioxide (CO <sub>2</sub> ).                                 |

## Section 11 - Toxicological Information

### Acute Effects

**Information on likely routes of exposure**

**Product Information** No acute toxicity information is available for this product

|            |                                    |
|------------|------------------------------------|
| Inhalation | Not an expected route of exposure. |
| Eyes       | Avoid contact with eyes.           |
| Skin       | Avoid contact with skin.           |
| Ingestion  | May be harmful if swallowed.       |

**Numerical measures of toxicity**

|                     |                   |
|---------------------|-------------------|
| (a) acute toxicity; |                   |
| Oral                | No data available |
| Dermal              | No data available |
| Inhalation          | No data available |

(b) skin corrosion/irritation; No data available

(c) serious eye damage/irritation; No data available

|                                        |                   |
|----------------------------------------|-------------------|
| (d) respiratory or skin sensitization; |                   |
| Respiratory                            | No data available |
| Skin                                   | No data available |

(e) germ cell mutagenicity; No data available

(f) carcinogenicity; No data available  
There are no known carcinogenic chemicals in this product

(g) reproductive toxicity; No data available

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; No data available

|               |                           |
|---------------|---------------------------|
| Target Organs | No information available. |
|---------------|---------------------------|

(j) aspiration hazard; No data available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

**Symptoms / effects, both acute and delayed**

Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting.

## **Section 12 - Ecological Information**

**Ecotoxicity**

**Aquatic ecotoxicity** Contains no substances known to be hazardous to the environment or that are not degradable in waste water treatment plants.

|                                      |                                                                                                                                                             |
|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Terrestrial ecotoxicity              | There is no data for this product                                                                                                                           |
| <b>Persistence and Degradability</b> |                                                                                                                                                             |
| Persistence                          | Miscible with water, Persistence is unlikely, based on information available.                                                                               |
| Bioaccumulative Potential            | Bioaccumulation is unlikely                                                                                                                                 |
| Mobility                             | The product is water soluble, and may spread in water systems. Will likely be mobile in the environment due to its water solubility. Highly mobile in soils |
| <b><u>Other adverse effects</u></b>  |                                                                                                                                                             |
| Endocrine Disruptor Information      | This product does not contain any known or suspected endocrine disruptors                                                                                   |
| Persistent Organic Pollutant         | This product does not contain any known or suspected substance                                                                                              |
| Ozone Depletion Potential            | This product does not contain any known or suspected substance                                                                                              |

## Section 13 - Disposal Considerations

### Waste treatment methods

|                                     |                                                                                                                                                                                                                                                                                                                               |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Waste from Residues/Unused Products | Do not allow into drains or watercourses or dispose of where ground or surface waters may be affected. Wastes, including emptied containers, are controlled wastes and should be disposed of in accordance with all federal, E.P.A., state and local regulations. Assure conformity with all applicable regulations.          |
| Contaminated Packaging              | Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.                                                                                      |
| Other Information                   | Disposal agencies or waste contractors must comply with the New Zealand Hazardous Substances (Disposal) Regulations . Waste codes should be assigned by the user based on the application for which the product was used. Do not flush to sewer. Can be landfilled or incinerated, when in compliance with local regulations. |

## Section 14 - Transport Information

### NZS 5433:2020

|                         |                           |
|-------------------------|---------------------------|
| UN-No                   | UN1224                    |
| Proper Shipping Name    | KETONES, LIQUID, N.O.S.   |
| Technical Shipping Name | tetrahydro-4H-pyran-4-one |
| Hazard Class            | 3                         |
| Packing Group           | III                       |

### IATA

|                         |                           |
|-------------------------|---------------------------|
| UN-No                   | UN1224                    |
| Proper Shipping Name    | KETONES, LIQUID, N.O.S.   |
| Technical Shipping Name | tetrahydro-4H-pyran-4-one |
| Hazard Class            | 3                         |
| Packing Group           | III                       |

### IMDG/IMO

|                                                                          |                                                                                                                         |
|--------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| UN-No                                                                    | UN1224                                                                                                                  |
| Proper Shipping Name                                                     | KETONES, LIQUID, N.O.S.                                                                                                 |
| Technical Shipping Name                                                  | tetrahydro-4H-pyran-4-one                                                                                               |
| Hazard Class                                                             | 3                                                                                                                       |
| Packing Group                                                            | III                                                                                                                     |
| Environmental hazards                                                    | No hazards identified                                                                                                   |
| Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code | Not applicable, packaged goods                                                                                          |
| Special Precautions                                                      | No special precautions required. Please refer to the applicable dangerous goods regulations for additional information. |
| Additional information                                                   | None known                                                                                                              |

## Section 15 - Regulatory Information

### Safety, health and environmental regulations/legislation specific for the substance or mixture

#### National Regulations

There are no applicable tolerable exposure limits or environmental exposure limits according to the EPA Controls for Hazardous Substances

#### Certified handlers, tracking and controlled substance license requirements

Certified handlers are required for some substances. This includes substances requiring a controlled substance license, and most explosives, vertebrates toxic agents, and certain fumigants. Acutely toxic substances which are a Category 1 or 2, such as pesticides also require Certified handlers. Please check the Health and Safety at Work Act 2015 for further information. Tracking is required for some highly hazardous substances. These substances need to be under the control of an appropriately trained person or appropriately secured. Please check the Health and Safety at Work Act 2015 for further information.

#### Prohibition or notification/licensing requirements

Shown below are details of specific prohibition/notifications or licencing requirements when they apply.

### International Regulations

**Ozone Depletion Potential** This product does not contain any known or suspected substance

**Persistent Organic Pollutant** This product does not contain any known or suspected substance

**Rotterdam Convention (PIC)** Not applicable

**Authorisation/Restrictions according to EU REACH** Not applicable

### International Inventories

New Zealand (NZIoC), Australia (AICS), Europe (EINECS/ELINCS/NLP), Korea (KECL), China (IECSC), Taiwan (TCSI), Japan (ENCS), Japan (ISHL), Canada (DSL/NDL), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component                 | CAS No     | NZIoC | AICS | EINECS    | ELINCS | NLP | KECL | IECSC | TCSI |
|---------------------------|------------|-------|------|-----------|--------|-----|------|-------|------|
| Tetrahydro-4H-pyran-4-one | 29943-42-8 | X     | -    | 249-967-2 | -      | -   | -    | X     | X    |

| Component | CAS No | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDL | PICCS | ISHL | ENCS |
|-----------|--------|------|-----------------------------------------------|-----|-----|-------|------|------|
|           |        |      |                                               |     |     |       |      |      |

|                           |            |   |   |   |   |   |   |   |
|---------------------------|------------|---|---|---|---|---|---|---|
| Tetrahydro-4H-pyran-4-one | 29943-42-8 | - | - | - | - | X | X | - |
|---------------------------|------------|---|---|---|---|---|---|---|

Legend: X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

## Section 16 - Other Information

This safety data sheet complies with the requirements of the EPA Hazardous Substances (Hazard Classification) Notice 2020 and WorkSafe New Zealand Regulations

### Legend

**NZIoC** - New Zealand Inventory of Chemicals  
**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory  
**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List  
**IECSC** - Chinese Inventory of Existing Chemical Substances  
**PICCS** - Philippines Inventory of Chemicals and Chemical Substances  
**TWA** - Time Weighted Average  
**IARC** - International Agency for Research on Cancer  
**NZS 5433:2020** - Transport of Dangerous Goods on Land  
**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association  
**MARPOL** - International Convention for the Prevention of Pollution from Ships  
**LD50** - Lethal Dose 50%  
**EC50** - Effective Concentration 50%  
**WEL** - Workplace Exposure Limit  
**DNEL** - Derived No Effect Level  
**POW** - Partition coefficient Octanol:Water  
**vPvB** - very Persistent, very Bioaccumulative  
**VOC** - (Volatile Organic Compound)

**AICS** - Australian Inventory of Chemical Substances  
**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances  
**ENCS** - Japanese Existing and New Chemical Substances  
**KECL** - Korean Existing and Evaluated Chemical Substances  
**CAS** - Chemical Abstracts Service  
**ACGIH** - American Conference of Governmental Industrial Hygienists  
**PNEC** - Predicted No Effect Concentration  
**OECD** - Organisation for Economic Co-operation and Development  
**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code  
**LC50** - Lethal Concentration 50%  
**ATE** - Acute Toxicity Estimate  
**RPE** - Respiratory Protective Equipment  
**NOEC** - No Observed Effect Concentration  
**BCF** - Bioconcentration factor  
**PBT** - Persistent, Bioaccumulative, Toxic

### Key literature references and sources for data

HSNO classifications provided in the New Zealand Chemical Classification Information Database (CCID).  
<https://echa.europa.eu/information-on-chemicals>  
 Suppliers safety data sheet, Chemadviser - LOLI, Merck index, RTECS  
 EPA Guide to classifying hazardous substances in New Zealand  
 EPA - Assigning a product to an existing HSNO approval guide

### Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.  
 Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.  
 First aid for chemical exposure, including the use of eye wash and safety showers.

**Revision Date** 13-Mar-2023  
**Revision Summary** Not applicable

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

## End of Safety Data Sheet